

Kinect v2 Depth Configuration

Recommended Settings & Analysis — E-Trike

2026-08-24 · Jetson AGX Orin · ROS 2 Humble · libfreenect2 0.2.0

Executive Summary

The Kinect v2 depth sensor works on Linux via [libfreenect2](#), which reverse-engineered Microsoft's proprietary USB protocol. The single most impactful improvement measured on this vehicle is the **depth processing pipeline**: switching from the CPU pipeline (5 Hz) to a CUDA pipeline reaches the sensor's native 30 Hz, and the **KDE variant** additionally improves ToF phase reconstruction.

This document gives the recommended production configuration, the measured evidence behind it, and the reasons for each setting.

Recommended Configuration

```
depth_pipeline: "cudakde"  
depth_bilateral_filter: true  
depth_edge_aware_filter: true  
depth_min_m: 0.3  
depth_max_m: 4.5
```

These map to the node parameters in `config/kinect_front.yaml` and `config/kinect_rear.yaml`, and are now actually applied to the device via `setConfiguration()` (previously dead parameters).

Measured Evidence

Test setup: front Kinect (serial 500076343042), fixed indoor scene, depth-only at 512×424, FPS via `ros2 topic hz`, pixel statistics via a dedicated subscriber over 20+ frames.

Pipeline	FPS	Valid px	Valid %	Depth std (m)
cpu	5.0	85,000	40%	1.42
cuda	30.0	98,200	45%	1.33
cudakde	29.3	98,700	45%	1.38

Interpretation

- **CPU is not real-time.** 5 Hz is unusable for obstacle detection on a moving vehicle.
- **CUDA variants reach 30 Hz** — the sensor's native rate — and recover 13,000 more valid depth pixels per frame than CPU.
- **KDE vs plain CUDA** are frame-rate-equal. KDE performs multi-hypothesis phase unwrapping with kernel-density estimation, yielding better outlier rejection. The valid-pixel count is marginally higher. This is why `cudakde` is recommended.

- The mean/range of valid depth is scene-dependent; the standard deviation column is indicative noise, not lab-grade (the scene shifted slightly between runs).

Why Each Setting

Pipeline: cudakde

The Kinect v2 does not transmit finished depth. Its ToF sensor measures phase at multiple modulation frequencies with nine correlation measurements; software reconstructs distance. Two libfreenect2 algorithms exist:

Algorithm	Source	Approach
Standard	<code>cuda_depth_packet_processor.cu</code>	Phase unwrapping via sensor P0/LUT tables; one hypothesis per pixel
KDE	<code>cuda_kde_depth_packet_processor.cu</code>	Multiple depth hypotheses per pixel; kernel-density estimation picks the most plausible; better outlier rejection and phase unwrapping

The KDE algorithm was developed specifically for Kinect v2 and reported better depth than both Microsoft’s SDK and the original algorithm.

Filters

Bilateral filter removes isolated “flying pixels” by comparing each pixel to its neighborhood. **Edge-aware filter** rejects unreliable ToF measurements at depth discontinuities (object silhouettes). Both are inexpensive on the GPU and reduce the two most common ToF artifacts. They default to enabled upstream; we keep them on.

Range clipping

Depth is clipped to $[0.3, 4.5]$ m inside the driver. Lowering the minimum to 0.3 m captures close obstacles relevant to a three-wheeler; the ToF sensor is accurate near-field. Keeping the maximum at 4.5 m avoids reporting garbage beyond the sensor’s physical envelope.

Warning: Direct sunlight physically overwhelms the IR ToF detector — no software setting can recover outdoor depth. The Kinect v2 is fundamentally a short-range indoor sensor.

Next Steps

1. **Set** `depth_pipeline: cudakde` in both config files (currently `auto`, which resolves to plain CUDA).
2. **Per-unit calibration** — the reference [iai_kinect2](#) tooling produces `data/<serial>/calib_{color,ir,pose,depth}.yaml`, correcting the 13-24 mm static depth offset and color distortion. This is the largest remaining accuracy gain.
3. **Downstream filtering** — a motion-aware temporal filter (blend only when the scene is static) plus a separate confidence channel, rather than aggressive hole-filling, is safe for an autonomous vehicle.

4. **Future** — SelfReDepth (2024) or GIGA-ToF (ICCV 2025) neural restoration on the Orin are research-stage options; treat neural depth as perception assistance, not ground truth.

Generated by Typst from the E-Trike Kinect depth analysis. Companion: depth-analysis.md (this folder)